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# GOTTA Prototype: Known-Asteroid Extraction and Statistical Performance Evaluation

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Abstract The GOTTA Prototype is a wide-field, high-cadence pathfinder for the full Global Open Transient Telescope Array. Its repeated survey imaging is expected to contain numerous detections of Solar System small bodies and provides an opportunity to validate moving-object processing before the full array is deployed. We present a known-asteroid recovery workflow for GOTTA Prototype photometric catalogs and use cataloged minor planets as a controlled sample to evaluate the prototype performance. Applying this workflow to the 2025 January–June prototype data yields 56,230 detections of 16,053 unique known asteroids over 73 UTC nights, including 77 near-Earth objects and 19 potentially hazardous asteroids. The recovered sample is dominated by main-belt asteroids. The observed-minus-predicted positional residual has a median separation of 0.220 arcsec and a two-dimensional RMS of 0.355 arcsec, demonstrating that the calibrated catalogs support sub-arcsecond known-object association. We further characterize the recovered sample in sky position, orbital class, aperture photometry, observing geometry, angular rate, and repeated temporal sampling. As a pilot light-curve application, selected GOTTA detections are combined with auxiliary 60 cm Schmidt observations, yielding 12 reliable rotation-period measurements and 12 additional tentative candidates requiring follow-up. These results provide a practical validation of the GOTTA Prototype moving-object workflow and establish a baseline for future known-object recovery, astrometric reporting, NEO follow-up support, asteroid light-curve analysis, and discovery-oriented processing with the full GOTTA array.

Key words: surveys — minor planets, asteroids: general — techniques: photometric — techniques: astrometric — methods: data analysis

## 1 INTRODUCTION

Near-Earth objects (NEOs) and potentially hazardous asteroids (PHAs) have long motivated systematic searches for small bodies in the Solar System. The Spaceguard Survey emphasized that the first step toward mitigating impact hazards is a comprehensive search for Earth-crossing asteroids and comets, followed by reliable orbit determination and long-term tracking (Morrison 1992). During the past three decades, wide-field CCD surveys have transformed this task from targeted searches into survey-scale operations. Programs such as Spacewatch, LINEAR, the Catalina Sky Survey, Pan-STARRS, and ATLAS have greatly increased the inventory of known minor planets and have established the basic observing and processing framework for modern moving-object surveys (e.g. Stokes et al. 2000; Larsen et al. 2007; Larson et al. 2003; Denneau et al. 2013; Tonry et al. 2018).

Moving-object science in time-domain surveys depends not only on single-image depth, but also on cadence, sky coverage, astrometric precision, and the ability to associate detections across time. A discovery-oriented pipeline generally starts from transient or non-stationary detections, forms same-night tracklets, and links tracklets obtained on different nights to determine preliminary orbits (Kubica et al. 2007; Denneau et al. 2013; Vereš & Chesley 2017). The same data stream also supports attribution and precovery of known objects, which are essential for orbit refinement, survey validation, and physical characterization. This distinction is particularly important for prototype surveys: even before a full discovery pipeline is finalized, the recovery of cataloged asteroids can be used to evaluate astrometric performance, photometric behavior, source extraction quality, and temporal sampling under real observing conditions.

Several recent and forthcoming surveys illustrate the range of strategies relevant to Solar System time-domain work. ATLAS was designed as a high-cadence all-sky system optimized for the discovery of dangerous near-Earth asteroids and other rapidly varying sources (Tonry et al. 2018). ZTF combines a 47 deg$^2$ camera with near-real-time reduction and alert distribution for transients, variables, and moving objects (Bellm et al. 2019; Patterson et al. 2019). The Rubin Observatory Legacy Survey of Space and Time will provide much deeper multi-epoch imaging and is expected to substantially expand the number of Solar System detections (Ivezić et al. 2019; Jones et al. 2018; Schwamb et al. 2023). In parallel, the NEO Surveyor mission will search in the infrared from space and is optimized for discovering and characterizing NEOs, especially objects difficult to observe from the ground (Mainzer et al. 2023). These facilities show that cadence, depth, wavelength, and sky coverage are complementary rather than interchangeable survey properties.

High-cadence survey data are also valuable for asteroid physical characterization. Time-series photometry constrains rotation periods and amplitudes, and multi-geometry light curves can be used to infer spin states and shapes through light-curve inversion (Kaasalainen & Torppa 2001; Kaasalainen et al. 2001; Warner et al. 2009). Sparse survey photometry further enlarges the statistical sample of asteroid light curves when sufficient temporal and phase-angle coverage is available. Accurate photometry and observing geometry are therefore necessary not only for detection and orbit association, but also for studying asteroid rotation, shape, surface properties, and activity. Phase-angle effects must also be treated explicitly when survey photometry is interpreted physically, for example through the $H, G_1, G_2$ phase-function formalism (Muinonen et al. 2010).

The SiTian/GOTTA program is designed to provide high-cadence, wide-field optical monitoring with a distributed array of telescopes (Liu et al. 2021). Its pathfinder, the Mini-SiTian Array at Xinglong Observatory, has already demonstrated the scientific value of high-cadence, multi-band observations for time-domain astronomy and has provided operational experience for the full SiTian/GOTTA system (Huang et al. 2025; Han et al. 2025). The Mini-SiTian Array consists of three 30 cm telescopes, has operated at Xinglong since 2022, and provides large-field photometric monitoring that informs both observation strategy and data processing for the full SiTian/GOTTA system. These pathfinder data and procedures are directly relevant to the development of Solar System analysis tools for future high-cadence survey operations.

For asteroid science specifically, Liu et al. (2025) developed an asteroid photometry and period-analysis pipeline using Mini-SiTian data from the f02 region. They derived rotation periods for 22 asteroids, including 14 objects without periods listed in ALCDEF, and found agreement with existing results for several objects with sufficiently dense photometric sampling. This work demonstrates the feasibility of asteroid photometry with SiTian pathfinder observations and provides an important motivation for building the corresponding known-object association and validation layer for the GOTTA Prototype.

The GOTTA Prototype extends this pathfinder context to a larger-aperture wide-field system. Mini-SiTian has shown that small pathfinder facilities can deliver useful asteroid light curves, while the GOTTA Prototype provides a testbed for survey-scale source catalogs, exposure geometry, and small-body database augmentation. The present work does not attempt to report unknown asteroid discoveries, measure formal completeness, or operate a full moving-object linking system. Instead, it focuses on known-object recovery, a necessary precursor to tracklet construction, moving-object alert filtering, orbit refinement, and light-curve preparation.

In this paper we present a known-asteroid extraction pipeline for GOTTA Prototype photometric catalogs. We use cataloged minor planets as a controlled sample to validate the end-to-end association between survey catalogs and known-object ephemerides. This approach allows us to evaluate astrometric residuals, photometric measurement behavior, angular-rate coverage, orbital-class distribution, and repeated-observation statistics. The recovered sample contains 56,230 matched detections of 16,053 unique known asteroids, including a small but useful subset of NEOs and PHAs. We also include a pilot light-curve analysis using auxiliary 60 cm Schmidt observations to demonstrate how the recovered known-object catalog can feed into an asteroid period-search workflow.

The paper is organized as follows. Section 2 describes the GOTTA Prototype data and the recovered known-asteroid sample. Section 3 presents the known-object association workflow and the period-analysis method used for the light-curve demonstration. Section 4 analyzes the recovered sample, evaluates the association performance, and presents the pilot light-curve results with auxiliary 60 cm Schmidt observations. Section 5 discusses the implications for future GOTTA moving-object processing and summarizes the conclusions.

## 2 OBSERVATIONS AND CATALOG DATA

### 2.1 GOTTA Prototype Survey Data

The observations analyzed in this work were obtained with the GOTTA Prototype during the original four-CMOS test campaign from 2025 January to June. The prototype is a 1 m-class wide-field Ring Schmidt system located at Xinglong Observatory and was operated during this stage with a single SDSS-like $g$ filter. The optical design, detector layout, and observing configuration are described in detail by Zhengyang Li et al. (in preparation). Only the aspects relevant to known-asteroid recovery are summarized here.

The present analysis uses calibrated source catalogs produced by the GOTTA Prototype basic data-reduction pipeline. That pipeline performs calibration-frame construction, image preprocessing, astrometric calibration, photometric calibration, and source extraction, as described by Niu Li et al. (in preparation). The astrometric solution is tied to Gaia DR3 (Gaia Collaboration et al. 2023), and the photometric catalog provides source positions, aperture photometry, Kron-like photometry, photometric uncertainties, and quality flags for each exposure. Related CMOS wide-field reduction strategies have also been developed for Mini-SiTian (Xiao et al. 2025). For known-asteroid recovery, the most important quantities are the WCS, the exposure time and observing epoch, the measured source coordinates, and the aperture-photometry measurements.

The companion reduction analysis shows that the calibrated catalogs are sufficiently accurate for the sub-arcsecond source associations required in this work. Stellar astrometric residuals are much smaller than the matching scale used here, while the photometric catalog includes aperture-corrected measurements in a sequence of circular apertures. Asteroid detections are generally fainter than the stars used for calibration and may be affected by motion during the exposure. Their observed-minus-predicted residuals therefore include additional contributions from centroiding uncertainty, ephemeris uncertainty, timing uncertainty, and possible trailing.

### 2.2 Recovered Known-Asteroid Sample

The recovered sample consists of accepted associations between predicted positions of cataloged asteroids and calibrated GOTTA source detections. Each association records the measured source

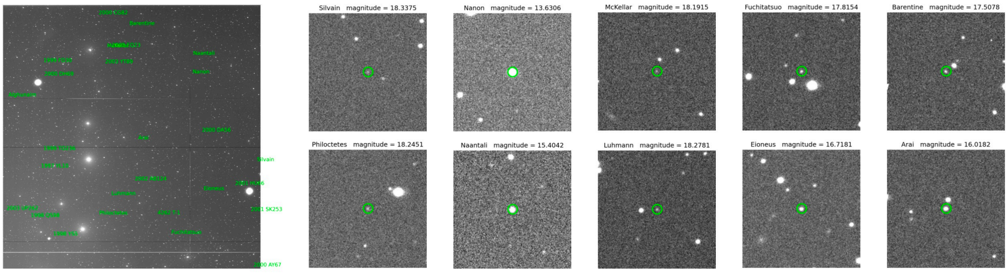


Fig. 1: Representative known-asteroid detections and image cutouts from the pilot-survey material. Green labels and circles identify recovered known asteroids. The figure illustrates the type of image-level detections represented by the accepted associations analyzed in this work.

position and photometry, the predicted ephemeris, the observing epoch, and auxiliary orbital and observing-geometry quantities. The final sample contains 56,230 detections of 16,053 unique known asteroids over 73 UTC nights. It is dominated by main-belt asteroids, with smaller contributions from Mars-crossing asteroids, Jupiter Trojans, NEOs, and PHAs. A subset of objects with repeated detections is used in the pilot period analysis described in Sections 3.4 and 4.4.

Figure 1 shows representative image cutouts of recovered asteroids. These examples illustrate the type of source detections included in the statistical analysis; all quantitative results below are derived from the full recovered sample.

## 3 METHODS

### 3.1 Known-Object Association Strategy

The goal of the pipeline is to identify detections of cataloged asteroids in calibrated GOTTA Prototype source catalogs. The procedure is designed as a known-object association workflow rather than a discovery-oriented linking pipeline. For each exposure, the pipeline reads the calibrated photometric catalog and the corresponding FITS header. The WCS solution is used to determine the sky footprint of the detector, while the exposure midpoint is computed from the start time and exposure duration. Known-asteroid candidates are predicted for this epoch and for the Xinglong observer location. Objects outside the padded field region or outside the CCD footprint are removed before cross-matching. We do not apply a cut on the predicted ephemeris magnitude, because activity, rotational modulation, phase-function uncertainty, and inaccurate absolute magnitudes can cause the measured brightness to differ from the nominal prediction. The remaining predictions are associated with catalog detections using an angular nearest-neighbor criterion. Likely stationary-source contaminants are rejected with a Gaia DR3 cross-check before the recovered detections are merged with source photometry and augmented with orbit and observing-geometry quantities from external small-body databases.

The statistics presented in this paper are produced by applying the association workflow to the accumulated prototype data set in batch mode. In parallel, we have also implemented a single-night processing version for future survey operations. The two modes share the same core steps—ephemeris prediction, footprint filtering, catalog association, Gaia-based stationary-source rejection, and small-body augmentation—but differ in their execution granularity. The batch mode is used here to obtain homogeneous statistics over the full data set, whereas the single-night mode is designed to produce nightly known-object products during routine operations. Figure 2 illustrates this single-night implementation. The present analysis uses only recovered detections and is therefore appropriate for evaluating the quality and content of the known-object sample, but not for measuring a formal detection efficiency, which would require the full set of predicted but unrecovered objects.

### 3.2 Ephemeris Prediction and Exposure Geometry

For each cataloged asteroid, the ephemeris calculation starts from osculating orbital elements provided by a minor-planet orbital-element catalog, such as the Lowell Observatory astorb database or the standard MPC small-body orbit records. The elements specify the heliocentric orbit at a reference epoch. In the two-body approximation, the mean anomaly is advanced from the reference epoch to the exposure midpoint using

$$M(t) = M_0 + n(t - t_0), \tag{1}$$

where $n = \sqrt{\mu_\odot / a^3}$ is the mean motion. Kepler's equation,

$$M = E - e \sin E, \tag{2}$$

is then solved for the eccentric anomaly $E$. The position in the orbital plane is

$$x' = a(\cos E - e), \qquad y' = a\sqrt{1 - e^2} \sin E. \tag{3}$$

This vector is rotated into the heliocentric ecliptic frame using the argument of perihelion, inclination, and longitude of the ascending node. The topocentric direction is obtained by subtracting the observer position at the same epoch and transforming the resulting vector into the equatorial frame. The apparent right ascension and declination follow from this topocentric line-of-sight vector.

This two-body description summarizes the physical basis of the screening calculation. In practice, high-precision apparent ephemerides also require time-scale handling, light-time correction, observer location, and perturbations from major Solar System bodies. The fast screening step therefore uses orbital elements from a minor-planet orbit catalog, while the final accepted detections are augmented with database and observer-dependent quantities from the JPL Small-Body Database[1] and Horizons[2] systems. The Minor Planet Center[3] provides the broader astrometric-observation infrastructure and reporting context for Solar System small-body work. The Lowell Observatory astorb database[4] is used as an orbit-catalog resource for fast screening.

The exposure footprint is computed directly from the WCS of each calibrated catalog. The detector corners are projected from pixel coordinates to the sky, and the search radius is initialized

[1] https://ssd.jpl.nasa.gov/tools/sbdb_lookup.html
[2] https://ssd.jpl.nasa.gov/horizons/
[3] https://minorplanetcenter.net/
[4] https://asteroid.lowell.edu/main/astorb/

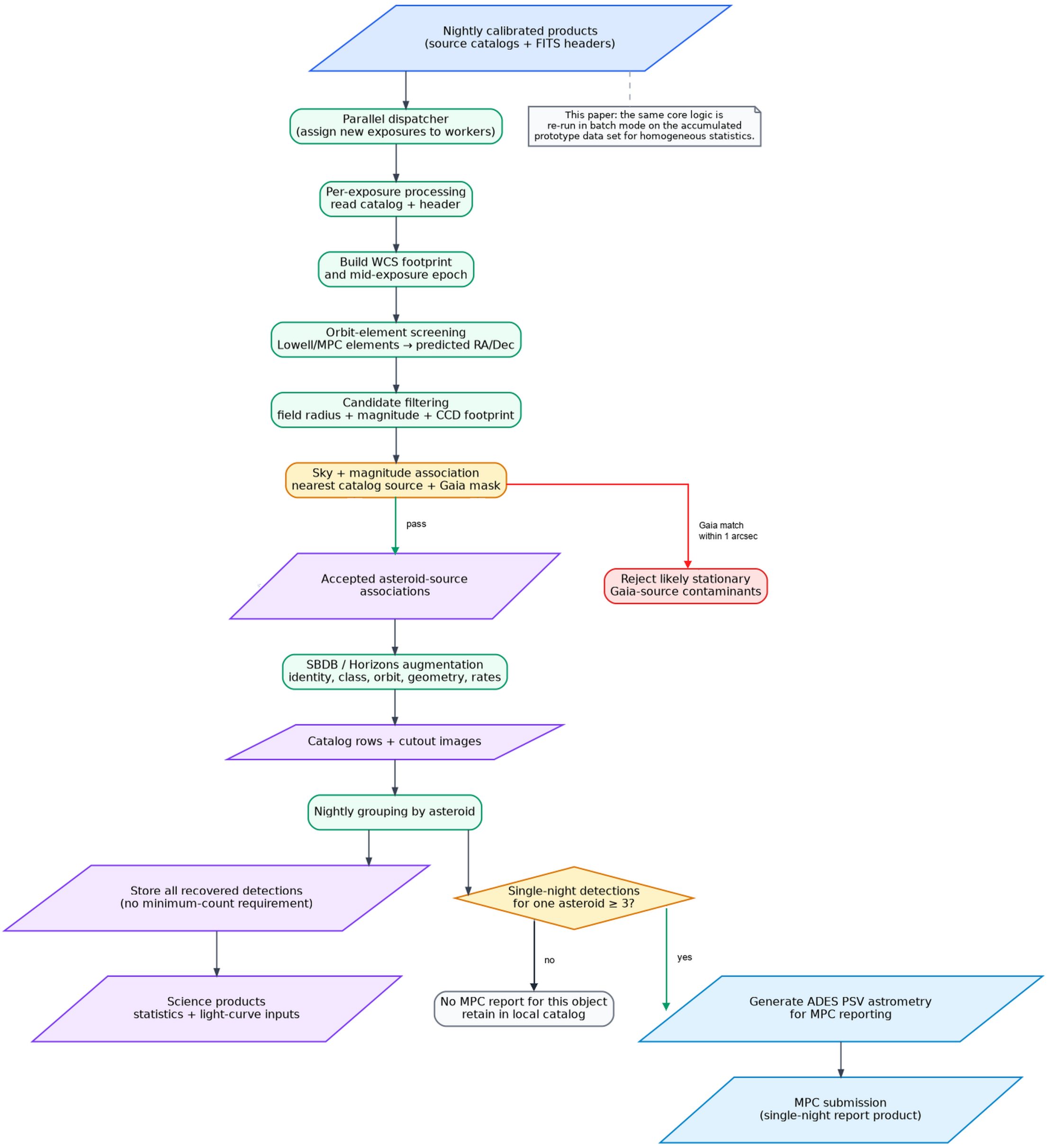


Fig. 2: Single-night workflow of the known-asteroid processing system. The batch statistics presented in this paper are produced by applying the same core association logic to the accumulated prototype data set. In the operational single-night mode, calibrated L2 source catalogs are dispatched to per-frame processing, matched detections are stored in the object-history catalog, and cutout images are generated for inspection. The association step is based on sky position, while the predicted magnitude is retained only for diagnostics. All accepted detections enter the matched-object history, while only objects with at least three detections in a night are selected for ADES PSV generation and MPC test submission.

from the maximum angular distance between the field center and the projected corners, with an additional padding term to avoid edge losses. The exposure midpoint is

$$t_{\rm mid} = t_{\rm start} + \frac{1}{2} t_{\rm exp}. \tag{4}$$

This midpoint is used for ephemeris evaluation because even modest sky-plane motion can lead to measurable offsets if the start time is used instead. After the initial sky-coordinate query, predicted positions are transformed back to detector coordinates, and only candidates falling within the valid pixel range are retained.

### 3.3 Catalog Matching and Small-Body Augmentation

The filtered predictions are then associated with the GOTTA source catalog. For each predicted position, the nearest catalog source is identified within a matching radius of 1 arcsec. The predicted ephemeris magnitude is retained for diagnostic comparison but is not used as an acceptance criterion. Small-angle residuals are reported in terms of $\Delta\alpha\cos\delta$, $\Delta\delta$, and their two-dimensional separation after the accepted matches have passed the Gaia stationary-source mask.

A Gaia-based stationary-source mask is applied after the asteroid-source association. A predicted asteroid position may occasionally fall close to a field star, in which case the nearest-neighbor source in the GOTTA catalog can be the stationary star rather than the moving object. To suppress this contamination, every accepted asteroid-source association is cross-matched with Gaia DR3 (Gaia Collaboration et al. 2023). If the same GOTTA source is also matched to a Gaia source within 1 arcsec, the association is rejected from the asteroid sample. This simple mask removes obvious stellar contaminants before the astrometric residuals are analyzed as a function of brightness and sky-plane angular rate.

For each accepted association, the measured catalog quantities are merged with the predicted ephemeris. The photometric measurements include aperture magnitudes and their uncertainties; Kron-like measurements are retained for comparison but are not used as the primary photometric statistic in the final analysis. The predicted asteroid magnitude is used only as a consistency diagnostic, because it is not necessarily defined in the same photometric system as the GOTTA $g$-band measurement and is also affected by asteroid color, phase behavior, and rotational variability.

The matched detections are further associated with small-body database entries to obtain object identifiers, orbit classes, orbital elements, and observer-dependent geometry. The latter includes heliocentric distance, topocentric distance, phase angle, and sky-plane angular rates. These quantities are used in the statistical analysis below to interpret the recovered sample in terms of orbital population, observing geometry, and apparent motion.

### 3.4 Light-Curve Period Analysis

Asteroid light curves are an important downstream product of moving-object survey processing because they constrain rotation periods, amplitudes, and basic shape and spin-state information. The known-object recovery workflow provides the object identities, observing epochs, and calibrated photometry required to construct asteroid time series. However, the GOTTA Prototype sampling alone is too sparse for robust period determination for most recovered objects. We therefore use a selected set of auxiliary observations from a nearby 60 cm Schmidt telescope to test the period-search component of the pipeline.

For each selected target, light curves are constructed from the available GOTTA Prototype and 60 cm Schmidt measurements. Aperture, PSF, and Kron-like photometry are processed separately as independent checks. Measurements with missing values or obvious outliers are removed, and a light curve is retained for period analysis only if it contains at least 60 valid points. This threshold suppresses spurious periodogram peaks caused by sparse sampling and provides a minimum phase coverage for model comparison. Only data sets with negligible inter-night or inter-instrument zero-point offsets are used in the current demonstration.

Candidate periods are searched using three complementary methods. The Lomb–Scargle periodogram is used for unevenly sampled time series (Lomb 1976; Scargle 1982). Phase Dispersion Minimization provides an independent search that is less sensitive to sinusoidal assumptions and is therefore useful for asteroid light curves (Stellingwerf 1978). A frequency-domain diagnostic based on a Fourier transform of an interpolated, uniformly sampled light curve is also used as an auxiliary check. The initial search focuses on short-period candidates; when a double-peaked interpretation is favored, the adopted rotation period can be twice the initial single-period candidate.

Candidate periods from different methods and photometric measurements are grouped in relative-period space. Periods that agree within 10 per cent are assigned to the same cluster, and the median period of the largest cluster is used as the initial candidate period. If no repeated period cluster is found, the object is not assigned a robust period at this stage. This consistency requirement prevents the final result from depending on a single periodogram peak.

Because asteroid light curves often show double-peaked morphology, both $P$ and $2P$ are tested for each candidate period. The phase is computed as

$$\phi = \left[\frac{t - t_0}{P}\right] \bmod 1, \tag{5}$$

where $t_0$ is an arbitrary reference epoch. The folded light curve is fitted with a Fourier series,

$$m(\phi) = a_0 + \sum_{k=1}^{N} \left[a_k \cos(2\pi k\phi) + b_k \sin(2\pi k\phi)\right], \tag{6}$$

where $N \leq 3$ in the current implementation. The fit is performed with photometric-error weighting and iterative sigma clipping. The choice between $P$ and $2P$ is evaluated using the residuals, phase coverage, the morphology of the folded light curve, and the Bayesian Information Criterion (Schwarz 1978),

$$\mathrm{BIC} = n\ln(\sigma^2) + k\ln n, \tag{7}$$

where $n$ is the number of valid measurements, $k$ is the number of model parameters, and $\sigma$ is the residual scatter.

The final high-confidence solutions are selected using both automatic and visual criteria. We examine the agreement among independent period-search methods, the number and symmetry of peaks in the folded light curve, the relative strength of the second harmonic, phase coverage, and the residual improvement between $P$ and $2P$. Folded light curves that are poorly sampled or dominated by obvious aliases are retained only as tentative candidates.

Table 1: Statistics for the recovered GOTTA known-asteroid sample.

| Quantity | Value | Note |
|---|---|---|
| Accepted source–ephemeris matches | 56,230 | Recovered detections |
| Distinct known asteroids | 16,053 | Unique minor-planet identifiers |
| Exposure catalogs with accepted matches | 13,253 | Per-exposure catalog products |
| UTC nights represented | 73 | Recovered sample date coverage |
| Field groups represented | 588 | Grouped by pointing metadata |
| NEO-class objects | 77 | Identified from small-body metadata |
| PHA-class objects | 19 | Identified from small-body metadata |
| Median ephemeris magnitude | 18.2215 | Predicted brightness |
| Median $g_{\rm aper}$ | 18.1110 | Adopted aperture magnitude |
| Median $\sigma(g_{\rm aper})$ | 0.1239 mag | Adopted aperture uncertainty |
| Median aperture S/N proxy | 8.7621 | Flux/error in adopted aperture |
| Median observed-minus-predicted separation | 0.220 arcsec | Recovered detections |
| 84th-percentile observed-minus-predicted separation | 0.484 arcsec | Recovered detections |
| 2D residual RMS | 0.355 arcsec | RMS of the 2D separation |
| RMS in $\Delta\alpha\cos\delta$ | 0.250 arcsec | Coordinate residual |
| RMS in $\Delta\delta$ | 0.252 arcsec | Coordinate residual |
| Median angular rate | 29.276 arcsec h$^{-1}$ | Derived from augmented geometry |
| Median phase angle | 13.245 deg | Derived from augmented geometry |
| Associations without geometry augmentation | 107 | Excluded from geometry statistics |

## 4 RESULTS

### 4.1 Recovered Sample, Orbital Populations, and Observing Geometry

The recovery procedure yields 56,230 accepted associations corresponding to 16,053 distinct known asteroids. These detections span 73 UTC nights during the 2025 January–June prototype campaign. The size of this sample is sufficient to characterize the dominant orbital populations recovered by the survey and to examine how the association residuals depend on brightness and apparent motion. Table 1 summarizes the main statistics.

The orbit-class composition is shown in Table 2. Main-belt asteroids account for 51,058 detections and 14,674 unique objects. Outer main-belt asteroids, Jupiter Trojans, inner main-belt asteroids, and Mars-crossing asteroids make up most of the remaining sample. Classes with only a few detections are grouped as Other/unclassified; this group includes objects without a more specific dynamical class in the adopted small-body metadata. The NEO classes are represented by

Table 2: Orbit-class composition of the recovered known-asteroid sample. Separations are in arcsec. Classes with only a few detections are grouped into "Other/unclassified"; this group includes objects without a more specific dynamical class in the adopted small-body metadata.

| Code | Orbit class | Detections | Unique objects | Median $g_{\mathrm{aper}}$ | Median separation |
|---|---|---|---|---|---|
| MBA | Main-belt Asteroid | 51,058 | 14,674 | 18.123 | 0.221 |
| OMB | Outer Main-belt Asteroid | 1,970 | 522 | 17.893 | 0.187 |
| TJN | Jupiter Trojan | 1,362 | 317 | 18.005 | 0.207 |
| IMB | Inner Main-belt Asteroid | 897 | 290 | 18.315 | 0.257 |
| MCA | Mars-crossing Asteroid | 656 | 166 | 17.847 | 0.228 |
| AMO | Amor | 139 | 31 | 17.143 | 0.227 |
| APO | Apollo | 101 | 38 | 17.749 | 0.253 |
| TNO | TransNeptunian Object | 20 | 3 | 17.588 | 0.243 |
| ATE | Aten | 19 | 8 | 17.668 | 0.292 |
| Other | Other / unclassified | 8 | 4 | 18.223 | 0.218 |

Table 3: Five UTC nights with the largest numbers of recovered known-asteroid detections.

| UTC date | Detections | Unique objects | Exposure catalogs | Median $g_{\mathrm{aper}}$ | Median separation |
|---|---|---|---|---|---|
| 2025-02-05 | 3,385 | 3,362 | 353 | 18.739 | 0.187 |
| 2025-01-19 | 3,119 | 3,065 | 576 | 18.354 | 0.189 |
| 2025-03-05 | 2,657 | 2,639 | 372 | 18.439 | 0.220 |
| 2025-01-28 | 2,656 | 2,638 | 395 | 18.499 | 0.189 |
| 2025-01-29 | 2,645 | 2,621 | 307 | 18.498 | 0.201 |

Amor, Apollo, and Aten objects. The dominance of main-belt objects is expected because the survey footprint and limiting depth preferentially sample objects near ordinary main-belt opposition geometries. The smaller NEO and PHA subsets should therefore be viewed as validation samples for the recovery path rather than as indicators of discovery performance.

The recovery rate also varies substantially from night to night because of sky coverage, observing conditions, and the number of available exposure catalogs. The five most productive UTC nights are listed in Table 3.

The recovered detections are concentrated in the surveyed sky regions available during the prototype campaign, reflecting the combination of the survey footprint, seasonal visibility, and the instantaneous distribution of known minor planets. The orbital distribution confirms that the sample is dominated by main-belt objects, with smaller contributions from inner and outer main-belt asteroids, Jupiter Trojans, Mars-crossers, and NEOs. These distributions are shown in Figures 3 and 4.

The corresponding observing geometry is summarized in Figure 5. The median heliocentric distance is 2.581 AU, the median topocentric distance is 1.840 AU, and the median phase angle is 13.245 deg. These values are consistent with a sample dominated by ordinary main-belt observing geometries and explain why the main angular-rate distribution is relatively narrow. Although the

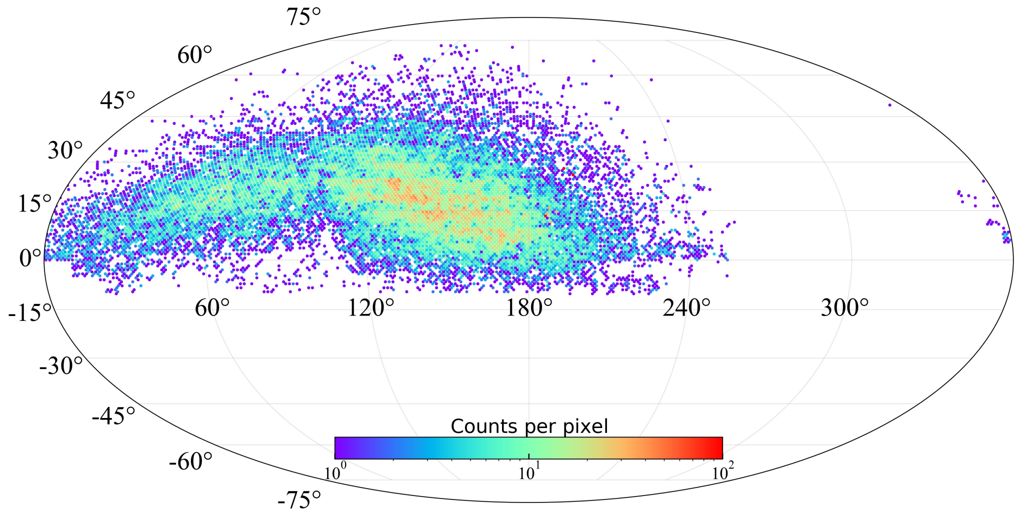


Fig. 3: Detection-level sky distribution of recovered known asteroids in equatorial coordinates, binned with HEALPix nside 64. The color scale is clipped at 100 detections per pixel for readability.

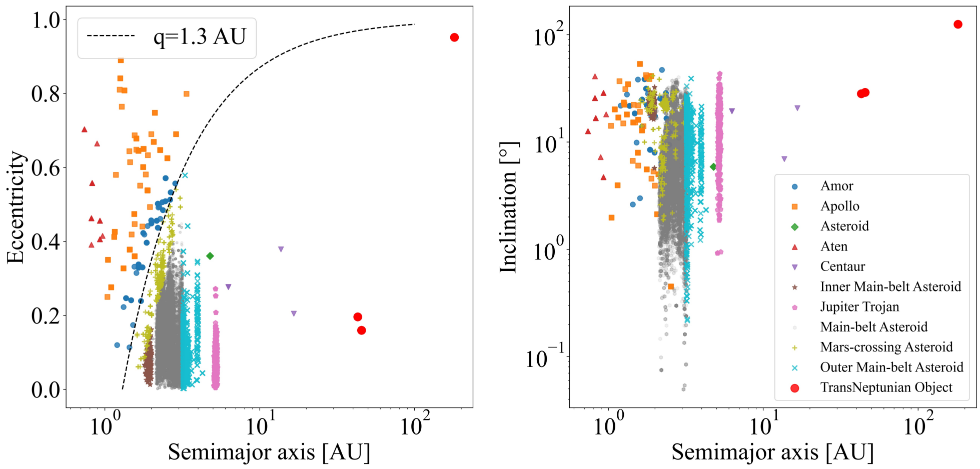


Fig. 4: Orbital-element distribution of the recovered known-asteroid sample. Points are colored by orbit class. The dashed curve in the semi-major-axis–eccentricity panel marks perihelion distance $q = 1.3$ AU, a conventional boundary for near-Earth objects.

sample is dominated by main-belt asteroids, the recovered NEO/PHA subset provides a useful stress test because these objects tend to have higher apparent rates and more variable observing geometry. The NEO subset contains 259 detections of 77 unique objects, with median $g_{\rm aper} = 17.525$ mag, median angular rate 111.643 arcsec $\rm h^{-1}$, and median residual separation 0.232 arcsec. The PHA subset contains 53 detections of 19 unique objects, with median $g_{\rm aper} = 17.660$ mag, median angular rate 132.424 arcsec $\rm h^{-1}$, and median residual separation 0.211 arcsec. These small subsets are not large enough for a formal population-level performance assessment, but they show that the current association path already covers higher-rate Solar System targets.

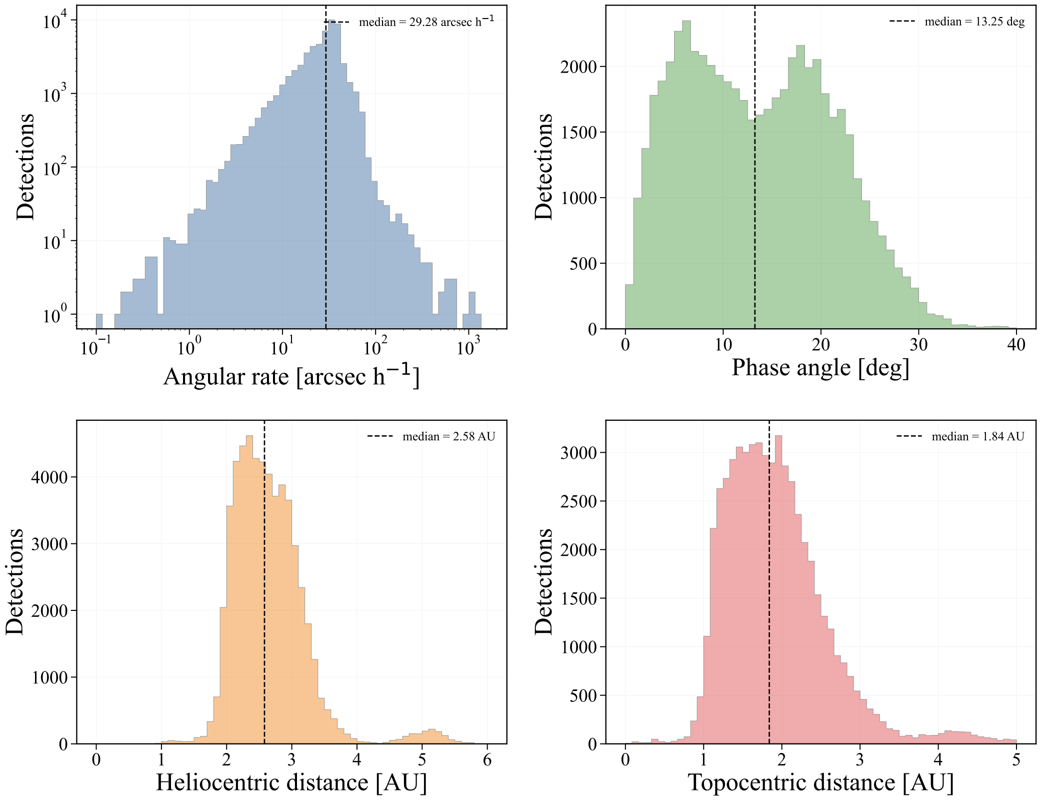


Fig. 5: Observing-geometry diagnostics for recovered known asteroids with successful geometry augmentation. The panels show sky-plane angular rate, phase angle, heliocentric distance, and topocentric distance. Dashed vertical lines mark the median values, which are listed in the legends.

Having established that the recovered catalog is dominated by main-belt objects observed at ordinary main-belt geometries, we next examine whether the measured brightness and photometric uncertainties are consistent with a sample of mostly faint single-exposure asteroid detections.

### 4.2 Photometric Properties of the Recovered Asteroids

For the photometric analysis we use the aperture magnitude corresponding to the optimal aperture defined by the basic GOTTA photometric pipeline. In that pipeline, aperture photometry is measured in a series of circular apertures, and the optimal aperture is determined from the signal-to-noise ratio of stellar sources as a function of aperture radius. A larger reference aperture is used for aperture correction, so that the cataloged aperture magnitudes are placed on a consistent photometric scale. We therefore do not re-derive an asteroid-specific optimal aperture in this work. Instead, we use the pipeline-defined optimal aperture as a homogeneous measurement for the recovered asteroid sample.

This choice provides a more homogeneous statistic than Kron-like photometry for the population-level analysis presented here. Kron apertures are adaptive and can respond differently to background noise, faint-source segmentation, and possible source elongation, whereas the adopted aperture magnitude provides a uniform measurement across the sample. Kron photom-

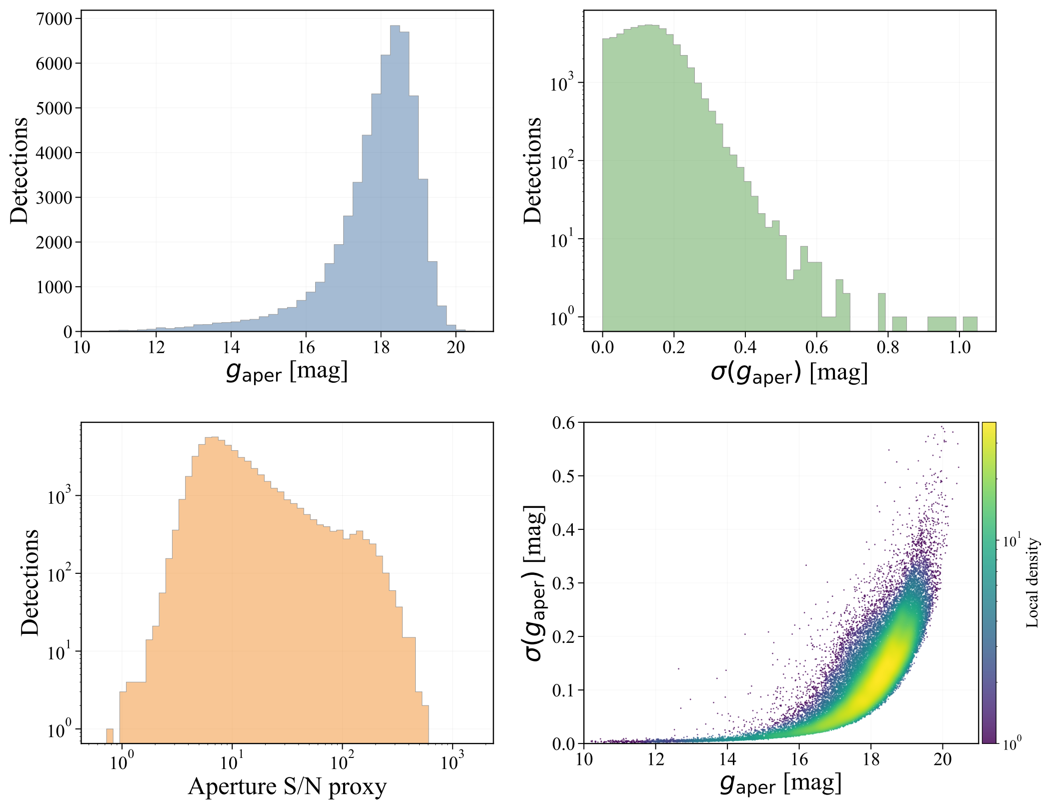


Fig. 6: Aperture-photometry diagnostics for recovered known asteroids. The adopted aperture follows the optimal aperture defined by the GOTTA basic photometric pipeline. The panels show the adopted aperture magnitude distribution, the corresponding magnitude-uncertainty distribution, the aperture S/N proxy, and the dependence of magnitude uncertainty on adopted aperture magnitude. The last panel is shown as a density map to emphasize the dominant distribution of faint asteroid detections.

etry is retained in the catalog for comparison, but it is not used as the primary quantity in the magnitude, uncertainty, and astrometric-residual diagnostics below.

We denote the adopted aperture magnitude as $g_{\rm aper}$ below. Most recovered asteroids lie close to the faint end of routine single-exposure catalog measurements rather than in the high signal-to-noise regime used for stellar calibration. The uncertainty–magnitude relation increases toward the faint end, as expected from decreasing source signal-to-noise ratio. The difference between predicted ephemeris magnitude and measured GOTTA $g$-band aperture magnitude is used only as a consistency diagnostic, because the two quantities are not necessarily defined in the same photometric system and may also differ because of asteroid color, phase behavior, rotation, shape, and ephemeris-magnitude uncertainties. The corresponding photometric distributions are summarized in Figure 6.

Because the photometric uncertainty rises rapidly toward the faint end, the same magnitude dependence is expected to appear in the astrometric residuals through centroiding uncertainty.

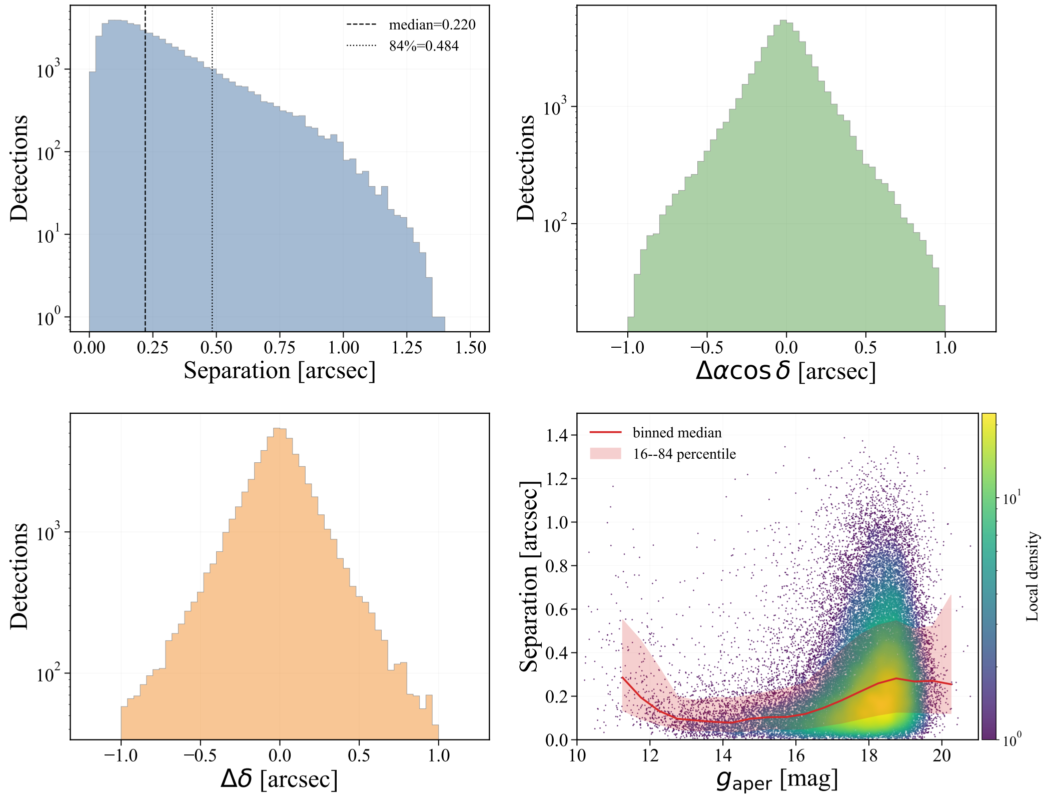


Fig. 7: Astrometric residual diagnostics. The panels show the observed-minus-predicted separation, $\Delta\alpha\cos\delta$, $\Delta\delta$, and separation as a function of the adopted aperture magnitude $g_{\rm aper}$. In the lower-right panel, the solid curve and shaded region indicate the binned median and 16th–84th percentile range in magnitude bins.

## 4.3 Astrometric Residuals and Apparent Motion

The residual distribution is strongly concentrated below the adopted matching scale, with a median separation of 0.220 arcsec and an 84th percentile of 0.484 arcsec. The two-dimensional RMS is 0.355 arcsec, and the RMS values of $\Delta\alpha\cos\delta$ and $\Delta\delta$ are 0.250 arcsec and 0.252 arcsec, respectively. The coordinate residuals are centered close to zero, indicating that the accepted associations do not show a large global offset between the predicted ephemerides and the calibrated source positions. When the residuals are grouped by adopted aperture magnitude, the median separation increases from about 0.09–0.11 arcsec for relatively bright detections to about 0.27 arcsec near $g_{\rm aper} \simeq 18$–19 mag. This trend indicates that source centroiding and signal-to-noise ratio dominate the residual budget for most recovered asteroids.

The dependence on sky-plane angular rate is weaker over the main part of the sample. Most recovered objects have moderate rates, with a median of 29.276 arcsec h$^{-1}$. In this regime, the median separation remains close to the full-sample value. The sparsely populated high-rate tail shows larger scatter and should be interpreted with caution because it contains few detections and may be more affected by trailing, centroiding biases, or matching ambiguity.

The magnitude and rate dependences are shown in Figures 7 and 8, where the binned medians summarize the main trends without introducing additional binned tables.

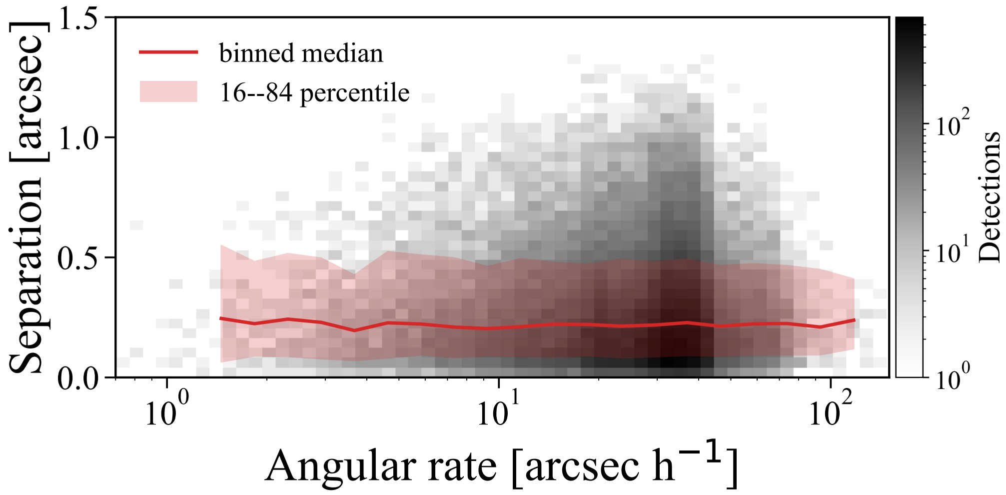


Fig. 8: Observed-minus-predicted separation as a function of sky-plane angular rate. The gray density map shows the dominant range of the recovered sample, while the red curve and shaded region show the binned median and the 16th–84th percentile range in logarithmic rate bins. Most detections lie at moderate angular rates, where the residual distribution remains close to the full-sample behavior. A small number of higher-rate detections are outside the displayed range for clarity; they are included in the full-sample summary statistics quoted in the text.

These residuals should be interpreted as association-level astrometric diagnostics rather than as a direct measurement of orbit improvement. For most well-observed main-belt asteroids, the predicted ephemeris uncertainty is expected to be smaller than the centroiding uncertainty of the faint GOTTA detections, so the observed-minus-predicted residual mainly traces the positional precision of the recovered survey source. However, no orbit refitting is performed in this work. The results therefore show that the prototype catalogs can provide sub-arcsecond asteroid positions suitable for known-object association and for future orbit-refinement experiments, but they do not by themselves quantify the improvement of minor-planet orbits.

The weak residual dependence on angular rate indicates that, for most detections, the sample is limited more by brightness and centroiding precision than by apparent motion. We therefore turn next to the observing geometry and revisit statistics that determine the time-domain usefulness of the recovered objects.

### 4.4 Temporal Sampling and Pilot Light-Curve Results

The scientific value of a high-cadence survey for asteroid work depends not only on the number of recovered objects, but also on how often individual objects are revisited. In the current sample the revisit statistics are highly non-uniform. The median number of detections per object is two, the mean is 3.50, and the 84th percentile is six. A small subset of objects has much denser sampling; the most frequently recovered object has 317 detections over nine distinct nights. The five most frequently recovered objects are listed in Table 4.

Table 4: Five most frequently recovered known asteroids in the GOTTA Prototype sample.

| Object ID | Name | Detections | Nights | First MJD | Last MJD | Baseline (days) | Median $g_{\rm aper}$ | Class |
|---|---|---|---|---|---|---|---|---|
| 214386 | 2005 MU1 | 317 | 9 | 60739.9 | 60818.7 | 78.824 | 18.141 | MBA |
| 76719 | 2000 JJ18 | 256 | 3 | 60738.8 | 60798.8 | 59.974 | 18.595 | MBA |
| 293 | Brasilia | 67 | 14 | 60709.9 | 60809.6 | 99.701 | 15.233 | MBA |
| 15751 | 1991 VN4 | 47 | 9 | 60714.8 | 60809.6 | 94.840 | 17.684 | MBA |
| 33629 | 1999 JK76 | 40 | 13 | 60709.9 | 60818.7 | 108.732 | 17.859 | MBA |

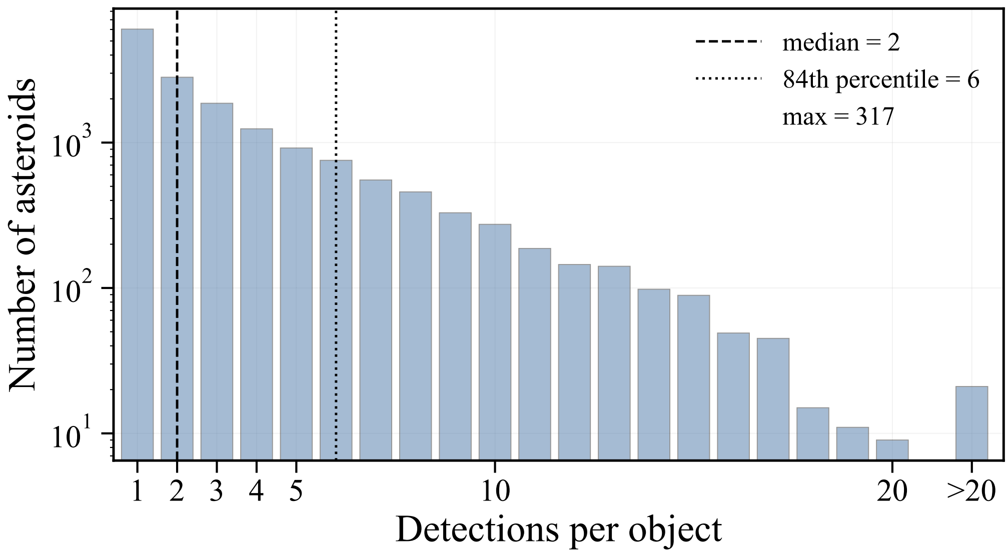


Fig. 9: Distribution of the number of recovered detections per object. The histogram emphasizes that most objects are recovered only a few times, while a small tail has tens to hundreds of detections. The median, 84th percentile, and maximum detection counts are indicated in the legend.

These repeated detections are useful for constructing sparse asteroid time series and for checking the internal consistency of the photometry. However, the prototype-only sampling is not sufficient for robust period determination for most objects. The distribution of detection counts per object is shown in Figure 9. This sampling limitation motivates the auxiliary light-curve experiment: the GOTTA detections provide object identities and part of the photometric time series, while the nearby 60 cm Schmidt telescope supplies denser temporal coverage for selected targets.

In the single-night operational workflow, objects recovered at least three times in one night are selected for ADES PSV generation. In the current batch sample, 16 object-night groups satisfy this criterion, corresponding to 15 unique objects. This statistic provides an operational estimate of how often the known-object pipeline produces MPC-reportable nightly products, although actual submission also requires additional quality control.

Applying the period-analysis procedure in Section 3.4 to the combined GOTTA Prototype and 60 cm Schmidt data yields 24 retained period-analysis candidates. Twelve objects satisfy the adopted high-confidence criteria, while the remaining twelve are classified as tentative candidates that require additional observations. Within the external period database queried in this work, none of the retained candidates has a matched reference period. We therefore treat these periods as validation-sample measurements rather than as a complete census of previously unknown asteroid rotation periods.

The reliable solutions are all classified as main-belt asteroids in the adopted small-body metadata. Their adopted rotation periods range from 2.84 to 6.14 hr, with a median of 4.71 hr. These values are typical of ordinary main-belt asteroid rotation periods and none approaches the approximate 2.2 hr spin barrier for strengthless rubble-pile asteroids. The pilot sample therefore provides a physically plausible check on the period-search workflow, although the present data do not yet support a population-level interpretation of amplitudes, shapes, or spin-state distributions. The tentative subset includes cases with weaker phase coverage, stronger alias ambiguity, or less secure folded-light-curve morphology, and should be treated as follow-up candidates.

The period-analysis validation sample is listed in Table 5. The first twelve entries are assigned reliable final quality flags, while the remaining entries are retained as tentative candidates requiring additional follow-up.

The reliable subset is used here to demonstrate that the recovered known-object catalog can feed into an asteroid period-search workflow when supplemented by denser auxiliary observations. Appendix Figures A.1 and A.2 show the folded light curves for the reliable and tentative subsets, respectively. The pilot analysis demonstrates the feasibility of the period-search pipeline, while also confirming that the GOTTA Prototype survey cadence alone is insufficient for routine period recovery for most objects.

Table 5: Period-analysis validation sample from the combined GOTTA Prototype and 60 cm Schmidt data.

| Object ID | $H$ (mag) | $N^{a}_{\mathrm{total}}$ | $N^{b}_{\mathrm{eff}}$ | Span (hr) | $P_{\mathrm{rot}}$ (hr) | $\Delta P$ (hr) | Model | Phot.[c] | MP[d] | Search[e] | Final[f] |
|---|---|---|---|---|---|---|---|---|---|---|---|
| 2175 | 14.42 | 796 | 671 | 123.4 | 4.8880 | 0.0010 | $P$ | Aper | 4 | Rel. | Rel. |
| 4479 | 13.26 | 1462 | 1454 | 195.9 | 3.0055 | 0.0197 | $P$ | Aper | 7 | Rel. | Rel. |
| 5424 | 13.57 | 757 | 722 | 99.9 | 5.4887 | 0.0014 | $2P$ | Kron | 3 | Good | Rel. |
| 7531 | 14.07 | 503 | 503 | 49.0 | 3.3375 | 0.0004 | $2P$ | PSF | 6 | Rel. | Rel. |
| 12148 | 14.50 | 459 | 457 | 29.5 | 4.5248 | 0.0104 | $2P$ | Aper | 5 | Rel. | Rel. |
| 19376 | 15.18 | 1093 | 1066 | 196.5 | 5.4896 | 0.0001 | $P$ | Aper | 6 | Rel. | Rel. |
| 22542 | 15.97 | 229 | 190 | 23.9 | 2.8439 | 0.0070 | $P$ | Kron | 2 | Poss. | Rel. |
| 25745 | 13.15 | 1541 | 1540 | 170.8 | 5.5143 | $< 0.0001$ | $2P$ | PSF | 6 | Rel. | Rel. |
| 46082 | 15.41 | 1211 | 1181 | 171.6 | 5.6907 | 0.0013 | $P$ | PSF | 6 | Rel. | Rel. |
| 72918 | 16.66 | 1473 | 1466 | 221.9 | 6.1385 | 0.0004 | $P$ | PSF | 6 | Rel. | Rel. |
| 79589 | 16.25 | 443 | 443 | 50.2 | 3.3634 | 0.0098 | $P$ | Aper | 9 | Rel. | Rel. |
| 97776 | 15.31 | 588 | 587 | 78.1 | 3.5371 | 0.0417 | $2P$ | PSF | 6 | Rel. | Rel. |
| 7558 | 13.45 | 438 | 392 | 51.3 | 4.1124 | 0.0019 | $P$ | PSF | – | Rel. | Tent. |
| 11382 | 15.46 | 615 | 591 | 53.1 | 15.2840 | 0.0273 | $2P$ | Aper | 6 | Rel. | Tent. |
| 17341 | 16.58 | 191 | 189 | 25.9 | 5.1682 | 0.0340 | $2P$ | PSF | 4 | Rel. | Tent. |
| 28036 | 13.42 | 226 | 222 | 4.9 | 4.4503 | 0.1598 | $P$ | PSF | 4 | Rel. | Tent. |
| 44099 | 16.02 | 216 | 209 | 4.8 | 3.5262 | 0.1210 | $2P$ | PSF | 6 | Rel. | Tent. |
| 50292 | 16.29 | 522 | 496 | 53.4 | 2.5186 | 0.0015 | $P$ | Aper | 4 | Rel. | Tent. |
| 51965 | 13.00 | 243 | 207 | 5.5 | 6.6819 | 0.0837 | $P$ | PSF | 6 | Rel. | Tent. |
| 54005 | 14.54 | 810 | 760 | 124.6 | 3.5362 | 0.0936 | $P$ | PSF | 4 | Rel. | Tent. |
| 57499 | 14.83 | 206 | 204 | 24.0 | 6.1133 | 0.0771 | $P$ | PSF | 4 | Rel. | Tent. |
| 68679 | 14.56 | 244 | 243 | 4.7 | 6.1144 | 0.1479 | $2P$ | Aper | 4 | Rel. | Tent. |
| 171725 | 16.69 | 286 | 276 | 26.3 | 5.0411 | 0.0118 | $2P$ | PSF | 5 | Rel. | Tent. |
| 750849 | 15.94 | 215 | 210 | 4.8 | 4.8756 | 0.1409 | $2P$ | Kron | 8 | Rel. | Tent. |

[a] Total number of photometric measurements available for the period-analysis pipeline.

[b] Number of measurements retained after quality cuts and outlier rejection.

[c] Photometric measurement used for the adopted solution: aperture (Aper), PSF, or Kron-like photometry.

[d] Number of period-search estimates in the adopted consistency cluster.

[e] Quality flag assigned from the agreement among the period-search methods.

[f] Final quality flag after folded-light-curve inspection: reliable (Rel.) or tentative (Tent.).

## 5 DISCUSSION AND CONCLUSIONS

The recovered known-asteroid sample demonstrates that calibrated GOTTA Prototype catalogs can support systematic association with cataloged Solar System objects. The main validation provided by this work is the closure of the known-object association workflow: calibrated source catalogs are combined with exposure geometry and mid-exposure timing, known-object ephemerides are filtered by the real CCD footprint, likely stationary-source contaminants are removed using Gaia cross-checks, accepted detections are matched in the source catalog, and the resulting associations are augmented with small-body database information. This workflow is a prerequisite for future moving-object products, including candidate-level bookkeeping, tracklet construction, and moving-object-aware transient filtering.

Known-object recovery is a useful validation mode because the predicted positions of cataloged asteroids are obtained independently of the GOTTA source extraction. The observed-minus-predicted residuals therefore test the combined effects of WCS quality, timing accuracy, centroiding, ephemeris uncertainty, and matching behavior. The median separation of 0.220 arcsec and the two-dimensional RMS of 0.355 arcsec show that the accepted associations are well localized on the scale required for known-object recovery. The increase of residual with fainter aperture magnitude indicates that centroiding precision and signal-to-noise ratio dominate the residual distribution for most of the sample. The weak dependence on angular rate over the main rate range suggests that trailing is not the primary limitation for the bulk of recovered objects, although the sparsely populated high-rate tail still requires dedicated analysis.

The residual analysis also clarifies what can and cannot be inferred about orbit refinement. For the majority of well-known main-belt objects, the ephemeris uncertainty is expected to be small compared with the centroiding uncertainty of faint GOTTA detections, so the observed-minus-predicted residuals provide a useful proxy for the astrometric quality of recovered moving sources. However, this work does not refit asteroid orbits using the GOTTA measurements. The results therefore do not quantify orbit improvement directly. They instead show that the prototype can deliver sub-arcsecond asteroid positions that are suitable for future astrometric reporting and orbit-refinement experiments, particularly for NEOs, PHAs, and objects with shorter observational arcs.

The photometric results should be interpreted as survey-catalog diagnostics rather than asteroid physical measurements. The adopted aperture magnitude provides a homogeneous brightness estimate, but the ephemeris magnitude is not a calibrated reference in the GOTTA $g$ band. Asteroid color, rotational modulation, phase-angle dependence, shape, and uncertainties in the magnitude model can all contribute to the difference between predicted and observed brightness. The pilot rotation-period analysis in Section 4.4 illustrates how the recovered known-object catalog can be used as the input layer for asteroid time-series work. The GOTTA-only sampling is generally too sparse for period determination, but the addition of 60 cm Schmidt observations allows the period-search method described in Section 3.4 to be tested on selected targets. The reliable periods obtained from the combined data should therefore be interpreted as a pipeline validation sample rather than as a complete rotational census.

Several limitations should be kept in mind when interpreting the present recovered sample. Because the catalog contains only accepted associations, it does not provide the denominator required for a formal recovery-efficiency estimate. The Gaia cross-check removes obvious stationary-source contaminants, but it does not by itself provide a complete false-association model. A quantitative false-match rate still requires the full set of predicted candidates, rejected associations, and original per-exposure source catalogs, together with random-shift or time-scrambling tests.

Fast-moving NEOs can also produce trailed or elongated images that are not optimally described by the point-source measurements used in the present catalog-level analysis. We are therefore developing a complementary machine-learning image-recognition module to identify and characterize trailed moving sources directly from image cutouts. This module will be particularly important for high-rate NEOs and will be integrated with the catalog-based known-object recovery workflow in future work.

For the future GOTTA array, the strength of the system will lie in the combination of large sky coverage and frequent temporal sampling. The prototype results show that the association and augmentation framework needed for known-object recovery can be implemented on GOTTA data. For the full GOTTA array, denser revisit patterns and multi-site coverage should increase the number of objects with sufficient phase coverage for period recovery, reducing the need for auxiliary observations for many targets. With candidate-level bookkeeping, repeated field visits, multi-band observations, and moving-object-aware scheduling, the same framework can be extended toward NEO monitoring, asteroid light-curve studies, and eventually discovery-oriented moving-object processing.

In summary, we have developed and tested a known-asteroid recovery workflow for GOTTA Prototype photometric catalogs. The recovered sample contains 56,230 detections of 16,053 unique known asteroids over 73 UTC nights, including small subsets of NEOs and PHAs. The sample is dominated by main-belt asteroids and provides statistically useful distributions in sky position, orbital class, aperture magnitude, angular rate, observing geometry, astrometric residual, and repeated temporal sampling. The sub-arcsecond observed-minus-predicted residuals demonstrate that the current catalog products are suitable for known-object association. The pilot light-curve analysis identifies 12 reliable rotation-period measurements from the combined GOTTA and 60 cm Schmidt validation sample, together with 12 tentative candidates requiring follow-up. This work establishes the first moving-object validation layer for the GOTTA Prototype and provides the basis for future candidate-level detection, tracklet construction, and asteroid light-curve analysis.

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Software: Astropy (Astropy Collaboration et al. 2013, 2018, 2022), NumPy (Harris et al. 2020), SciPy (Virtanen et al. 2020), Matplotlib (Hunter 2007), pandas (McKinney 2010), astroquery (Ginsburg et al. 2019), and healpy/HEALPix (Górski et al. 2005).

## Appendix A: LIGHT-CURVE VALIDATION FIGURES

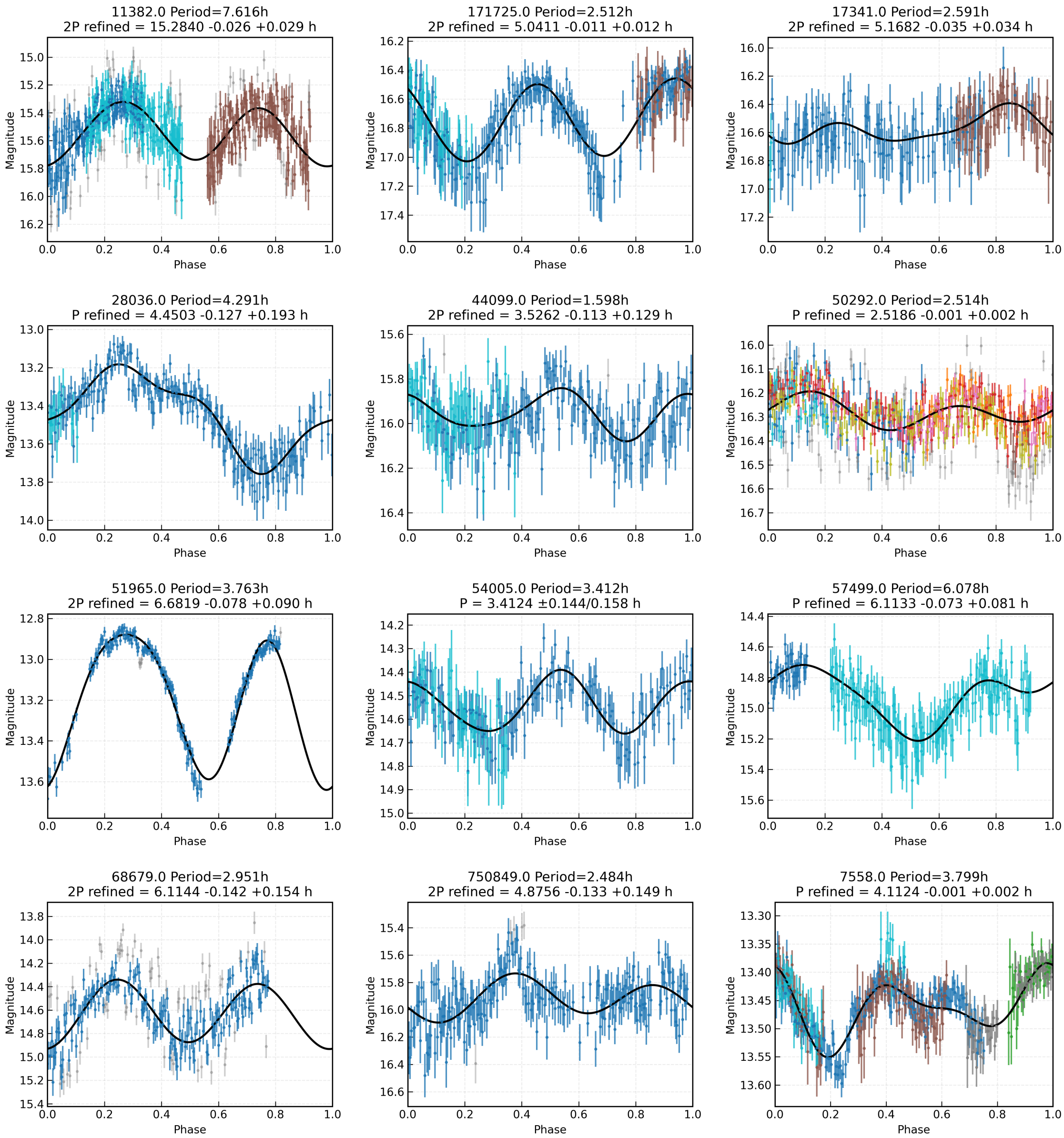


Fig. A.1: Phase-folded light curves for the 12 objects assigned reliable final quality flags in the updated combined GOTTA Prototype and 60 cm Schmidt validation sample. Colored points denote retained measurements from different data subsets, gray points mark rejected or lower-weighted measurements, and black curves show the adopted Fourier models. Each panel gives the object identifier and the candidate period information used in the current period analysis.

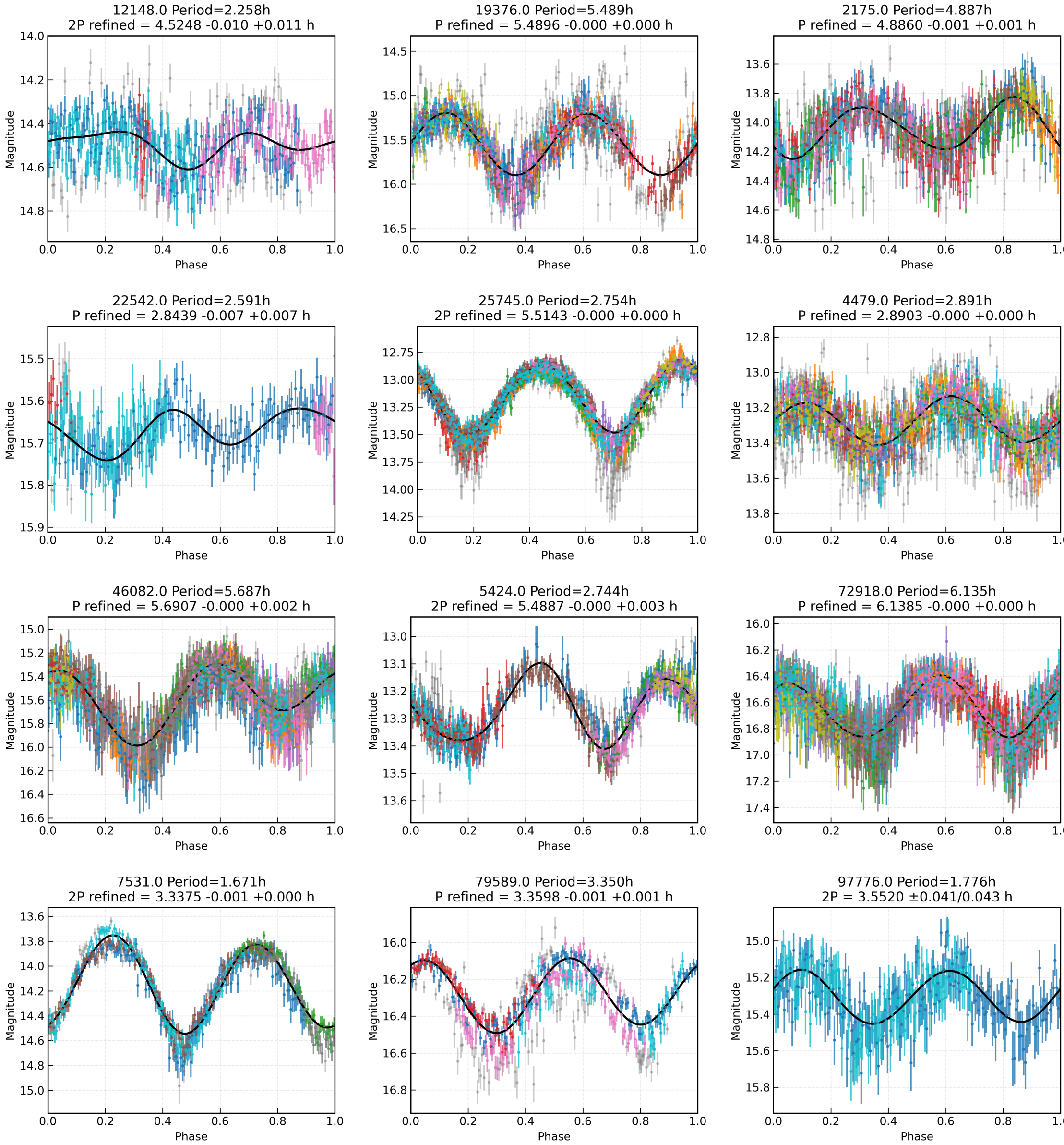


Fig. A.2: Same as Figure A.1, but for the 12 objects assigned tentative final quality flags. These solutions are retained in the validation sample but require additional observations for confirmation.

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